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Multiple Monoblock Aircraft Battery

Abstract

A battery unit for providing power to an aircraft may include one or more battery modules and a retaining structure. Each battery module comprises a monoblock and a control system coupled to the monoblock. The control system controls a switch that is actuatable to selectively engage or disengage a high power path circuit coupling the monoblock of each of the one or more battery modules to a battery power terminal connector of the battery unit. The retaining structure is configured to contain the one or more battery modules, and the retaining structure includes a monitoring system in communication with the control system of each of the one or more battery modules. The one or more battery modules are removable from the retaining structure by disconnecting at least the high power path circuit and a communication path between the control system and the monitoring system.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/556,590, filed Feb. 22, 2024, the entire contents thereof are herein incorporated by reference.

BACKGROUND

1. Field

[0002] Embodiments of the invention relate generally to battery modules, and more specifically to a multiple monoblock battery for providing power to an aircraft.

2. Related Art

[0003] Various solutions have been proposed for scalable battery modules that provide power and that feature safety protections from thermal events. U.S. Pat. Nos. 9,583,794 and 10,027,133 to Adrian et al. disclose a modular battery system having a plurality of battery sub-modules operably connected in parallel, and isolation and condition systems configured to isolate any one battery sub-module from the remaining battery sub-modules such that battery sub-modules can be individually added, removed, isolated and/or conditioned while other battery sub-modules continue to provide power. International Publication No. WO2018222546 and U.S. Pat. No. 10,903,464 to Huff disclose a multi-modular battery system having interchangeable individual battery modules that can be replaced and that communicate with each other using optical communication methods. Each battery module includes a plurality of cells. U.S. Patent Publication No. 2021/0203015 to Ahrens et al. discloses a modular battery device with one or more replaceable cell elements that are removable from the device without uninstalling the device or disrupting a power supply from the other cell elements in the device.

SUMMARY

[0004] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other aspects and advantages of the invention will be apparent from the following detailed description of the embodiments and the accompanying drawing figures.

[0005] The embodiments described herein relate to a battery unit including: one or more battery modules, each battery module including a monoblock and a control system coupled to the monoblock, the control system actuatable to selectively engage or disengage at least one first circuit coupled to the monoblock of each of the one or more battery modules; a retaining structure configured to contain the one or more battery modules and including a power connector that provides power from the one or more battery modules; a monitoring system that is in communication with the control system of each of the one or more battery modules; wherein: the at least one first circuit couples each monoblock of the one or more battery modules to the power connector and the monitoring system, each of the one or more battery modules is removable from the retaining structure by disconnecting at least the at least one first circuit and a communication path that couples the control system and the monitoring system, and the power connector provides power when the at least one first circuit of at least one of the one or more battery modules is engaged.

[0006] The embodiments described herein relate to a battery unit, further including a second circuit that couples the monoblock of each of the one or more battery modules to at least the monitoring system, and each of the one or more battery modules is removable from the retaining structure by further disconnecting a control power circuit that includes the second circuit that couples the control system of each of the one or more battery modules to the monitoring system.

[0007] The embodiments described herein relate to a battery unit, further including a second circuit that couples the monoblock of each of the one or more battery modules to at least the monitoring system, the second circuit including an aircraft signal connector that is configured to communicatively connect the monitoring system to an aircraft.

[0008] The embodiments described herein relate to a battery unit, wherein the control system is housed within an enclosure and the control system is thermally isolated from the monoblock.

[0009] The embodiments described herein relate to a battery unit, wherein the monoblock is removable from the control system.

[0010] The embodiments described herein relate to a battery unit, further including a manifold coupled to the one or more battery modules, the manifold configured to receive effluent gases produced within the monoblock of each of the one or more battery modules.

[0011] The embodiments described herein relate to a battery unit, wherein the control system of each of the one or more battery modules includes a temperature sensing device for sensing a temperature of the monoblock thereof, and the control system transmits data from the temperature sensing device to the monitoring system.

[0012] The embodiments described herein relate to a battery unit, wherein the monitoring system includes a fault management system that is in communication with the control system of each of the one or more battery modules to actuate the control system to selectively engage or disengage the at least one first circuit.

[0013] The embodiments described herein relate to a battery unit, wherein the fault management system stores data related to at least non-resettable fault events of each monoblock of the one or more battery modules retained in the retaining structure, and the monitoring system is in communication with the control system of each of the one or more battery modules to disengage the monoblock thereof having data stored in the fault management system related to a non-resettable fault event.

[0014] The embodiments described herein relate to a battery unit, further including a second circuit coupled to the monoblock of each of the one or more battery modules that includes means to test a voltage of the monoblock connected thereto.

[0015] The embodiments described herein relate to a battery unit configured to provide electrical power to an aircraft, the battery unit including: a retaining structure; a monitoring system; one or more battery modules retainable in the retaining structure, wherein each battery module includes: a monoblock including one or more battery cells, a first circuit that is coupled to the monoblock and engages at least the retaining structure and monitoring system, a second circuit that is coupled to the monoblock and engages at least the monitoring system, the second circuit including an aircraft signal connector that is configured to communicatively connect the monitoring system to the aircraft, and a control system that is actuatable to selectively engage or disengage the first circuit, wherein the control system is housed within an enclosure and is insulated from at least the monoblock of the battery module; wherein the control system of each of the one or more battery modules is in communication with the monitoring system via at least a communication path, the one or more battery modules are each removable from the retaining structure by disconnecting at least the first circuit, the second circuit, and the communication path, and the monoblock is removable from the control system.

[0016] The embodiments described herein relate to a battery unit, wherein the monoblock of each of the one or more battery modules includes a chamber configured to contain effluent gases produced by the one or more battery cells, and the battery unit further includes a manifold coupled

to the chamber of each of the one or more battery modules configured to receive effluent gases contained in the chamber of each monoblock.

[0017] The embodiments described herein relate to a battery unit, wherein the control system of each of the one or more battery modules includes a temperature sensing device for sensing a temperature of the monoblock thereof, and the control system transmits data from the temperature sensing device to the monitoring system.

[0018] The embodiments described herein relate to a battery unit, wherein the monitoring system includes a fault management system that is in communication with the control system of each of the one or more battery modules to actuate the control system to selectively engage or disengage the first circuit.

[0019] The embodiments described herein relate to a battery unit configured to provide electrical power to an aircraft, the battery unit including: a retaining structure having a power connector that is configured to be electrically connected to the aircraft; a monitoring system positioned within the retaining structure; one or more battery modules retained in the retaining structure and coupled to at least the monitoring system, wherein each battery module includes: a monoblock including one or more battery cells, a high power path circuit that is coupled to the monoblock and engages the power connector and the monitoring system, a low power path circuit that is coupled to the monoblock and engages at least the monitoring system at least when the high power path circuit of the battery module is disengaged, wherein the low power path circuit is a current-limited circuit, and a control system coupled to the monoblock and communicatively connected to the monitoring system, wherein the control system controls an electronic switch that is actuatable to selectively engage or disengage the high power path circuit, and the control system is housed within an enclosure and is insulated from at least the monoblock of the battery module; wherein the power connector of the retaining structure is configured to be electrically connected to the aircraft when the high power path circuit of at least one of the one or more battery modules is engaged, and the one or more battery modules are each removable from the retaining structure by disconnecting at least the high power path circuit, the low power path circuit, and a communication path between the control system of the one or more battery modules and the monitoring system.

[0020] The embodiments described herein relate to a battery unit, wherein the monoblock is removable from the control system.

[0021] The embodiments described herein relate to a battery unit, wherein the monoblock of each of the one or more battery modules includes a chamber configured to contain effluent gases produced by the one or more battery cells, and the battery unit further includes a manifold coupled to the chamber of each of the one or more battery modules configured to receive effluent gases contained in the chamber of each monoblock.

[0022] The embodiments described herein relate to a battery unit, wherein the control system of each of the one or more battery modules includes a temperature sensing device for sensing a temperature of the monoblock thereof, and the control system transmits data from the temperature sensing device to the monitoring system.

[0023] The embodiments described herein relate to a battery unit, wherein the monitoring system includes a fault management system that is in communication with the control system of each of the one or more battery modules to actuate the control system to selectively engage or disengage the high power path circuit that electrically connects the monoblock to at least the power connector and the monitoring system.

[0024] The embodiments described herein relate to a battery unit, wherein the low power path circuit includes an aircraft signal connector that is configured to communicatively connect the monitoring system to the aircraft.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0025] Embodiments of the invention are described in detail below with reference to the attached drawing figures, wherein:

[0026] FIG. 1 shows a perspective view of a multiple monoblock battery unit that includes three battery modules contained in a retaining structure and a vent manifold system, in an embodiment;

[0027] FIG. 2 shows an exploded perspective view of the multiple monoblock battery unit in FIG. 1, including the three battery modules, the retaining structure, the vent manifold system, a monitoring system coupled to a terminal wall of the retaining structure, a removable panel that supports an electrical bus system, and control system enclosures that are coupled to monoblocks to form the battery modules, in an embodiment;

[0028] FIG. 3 shows an exploded perspective view of the multiple monoblock battery unit in FIG. 1 from an opposite perspective as that shown in FIG. 2, and includes the three battery modules, the retaining structure, the vent manifold system, connectors and the monitoring system expanded from the terminal wall of the retaining structure, an electrical bus system coupled to the removable panel, and straps for moving and carrying the unit, in an embodiment;

[0029] FIG. 4 shows an exploded perspective view of one of the battery modules shown in FIG. 1, the battery module including a monoblock and a control system removably coupled thereto, the control system comprising a control PCB, an SSR PCB, and a control system enclosure, in an embodiment;

[0030] FIGS. 5 and 6 show a perspective view of the multiple monoblock battery unit in FIG. 1, and includes a secondary cover, in an embodiment;

[0031] FIG. 7 shows an elevation view of the multiple monoblock battery unit in FIG. 1, and includes tie-down rods connected to lugs of the vent manifold system, the tie-down rods configured to secure the multiple monoblock battery unit to the aircraft, in an embodiment; and

[0032] FIG. 8 shows a schematic representation of one of the battery modules communicatively coupled to a monitoring system of the multiple monoblock battery unit shown in FIG. 1, in an embodiment.

[0033] The drawing figures do not limit the invention to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the invention.

DETAILED DESCRIPTION

[0034] The following detailed description references the accompanying drawings that illustrate specific embodiments in which the invention can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized, and changes can be made without departing from the scope of the invention. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the invention is defined only by the appended claims, along with the full scope of the equivalents to which such claims are entitled.

[0035] In this description, references to “one embodiment,” “an embodiment,” or “embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment,” “an embodiment,” or “embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc. described in one embodiment may also be included in other embodiments but is not necessarily included. Thus, the technology can include a variety of combinations and/or integrations of the embodiments described herein.

[0036] In the embodiments described herein, a battery unit **100** may be a main aircraft battery unit that may be used for at least engine starts and as an emergency power supply. The battery unit **100** provides power to the aircraft or another system and comprises one or more battery modules **110**.

The battery unit **100** is configured to be electrically connected to an electrical power distribution system of the aircraft that may include an aircraft bus system to direct power to one or more aircraft systems and/or other electronic systems. Each battery module **110** may be removed from or added to the battery unit **100** or replaced or exchanged with another battery module **110**, such that the battery unit **100** may be scalable to be used for variety of applications and a useful lifetime of each of the components of the battery unit **100** may be maximized.

[0037] Each battery module **110** includes a monoblock **115** (as shown in FIG. **4**) containing one or more battery cells **120**. As used herein, the term battery module refers to a collection of cells (which may be in the form of a monoblock) coupled to electronic components, such as a control system; the term monoblock generally refers to a configuration of cells electrically coupled together to form a single (i.e., mono) block. The battery monoblocks **115** are each thermally insulated, and the monoblocks **115** are each segregated or isolated from the rest of the system, including the electronics, to prevent potential thermal events within the monoblock **115** from causing damage beyond the monoblock **115**. Each monoblock **115** may be configured to vent effluent gases collected therein to the environment when a threshold pressure and/or a temperature of an interior of the battery monoblock **115** of the battery module **110** is reached or exceeded.

[0038] In an embodiment, the battery unit **100** is configured such that failure of one monoblock **115** allows for continued operation the battery unit **100**, extending the lifetime of the battery unit **100**, eliminating disposal of electrical components that still have a useful life, for example, the control system, and lowering costs associated with replacement of the battery unit **100**. Failure of the monoblock **115** may include, for example, a thermal event, a short circuit, cell degradation, or another foreseeable issue or event.

[0039] As shown in FIG. **1**, the battery unit **100** comprises the one or more battery modules **110** contained in a retaining structure **121**. The battery unit **100** shown in FIG. **1** includes three battery modules **110**, but a battery unit **100** may include fewer or greater than three battery modules **110** without departing from the scope of the invention described herein. Each battery module **110** is removably mounted to the retaining structure **121**.

[0040] As shown in the exploded views of the battery unit **100** in FIGS. **2** and **3**, when each battery module **110** is mounted in the battery unit **100**, the battery module **110** is electrically connected to and in communication with a monitoring system comprising a monitoring system printed circuitry board (“monitoring system PCB”) **122** which may be adjacent to or incorporated in or coupled to an interior side of a terminal wall **123** of the retaining structure **121**. It is foreseeable that the monitoring system PCB **122** is a distinct and separate system that is not coupled to the retaining structure **121**. The monitoring system PCB **122** may also be referred to as a monitoring system **122**. The battery module **110** being “mounted” is defined herein as the battery module **110** being physically connected to or inserted in the retaining structure **121**, such as in FIG. **1**, and the mounted battery module **110** may be electrically engaged or connected to the battery unit **100** or electrically disengaged or disconnected from the battery unit **100**. The battery module **110** being “dismounted” is defined herein as the battery module **110** being physically disconnected or removed from the retaining structure **121** and electrically disengaged or disconnected from the battery unit **100**.

[0041] FIG. **4** shows an exploded view of a battery module **110**. Each battery module **110** comprises the monoblock **115** which includes the one or more battery cells **120**, such as rechargeable lithium-ion battery cells or other types of cells. In one embodiment, the monoblock **115** includes an array of 48 battery cells **120**, such as, for example, six groups of battery cells **120** connected in parallel, with each group including eight battery cells **120** connected in series. It is foreseeable that the number and configuration of the battery cells **120** may vary and such is within the scope of the embodiments described herein.

[0042] In one embodiment, each battery module **110** comprises battery cells **120** having the same or substantially similar chemistry as the battery cells **120** of the other battery modules **110** of the

battery unit **100**. The battery modules **110** of the battery unit **100** each have a similar energy capacity and/or power capacity as compared to the other battery modules **110** of the battery unit **100**. It is foreseeable that an integrated bi-directional DC to DC converter, such as that disclosed in U.S. Provisional Patent Application No. 63/493,939 to Tidball et al., and incorporated herein in its entirety by reference, may be used to convert a source of direct current (DC) from one voltage level to another to allow use of battery modules having different capacities and/or different chemistries. [0043] The retaining structure **121** may include a base **124** on which the battery modules **110** are positioned and wall members, including an upper wall **126** that is parallel to the base **124**, side walls **127**, and the terminal wall **123**. In a preferred embodiment, the base **124** is formed to support the battery modules **110** that are positioned side-by-side on the base **124**. In an embodiment, the base **124** may include a lip (not shown) circumventing a portion of the base **124** to contain one or more battery modules **110** in the retaining structure **121**. In an embodiment, the upper wall **126** of the retaining structure **121** may include a portion that extends down (not shown) to cover rearward ends **128** of the battery modules **110** in the battery unit **100** to retain the battery modules **110** therein. The upper wall **126** extends between the side walls **127**. The terminal wall **123** extends between the base **124**, the upper wall **126**, and the side walls **127**. As shown in an embodiment in FIGS. 5 and 6, a secondary cover **129** may be installed over and/or around the battery modules **110** to restrain and contain the battery modules **110**, especially during movement or vibration of the battery unit **100**, and to provide additional environmental protections, such as, for example, waterproofness. The secondary cover **129** may overlap or be coupled to a portion of the retaining structure **121** and/or the rearward ends **128** of the battery modules **110**.

[0044] The retaining structure **121** is formed such that an internal width of the retaining structure **121** is approximately a total width of the one or more battery modules **110** positioned therein, and the internal height of the retaining structure **121** is approximately a height of the battery modules **110**. Each battery module **110** extends lengthwise in the retaining structure **121** from a removable panel **151**, described herein, to the opposite end of the base **124**. In one embodiment, the side walls **127** are structural members that provide support for the battery modules **110**. The side walls **127** of the retaining structure **121** may or may not extend to cover at least a portion of a lateral wall **130** of the battery module **110** positioned lengthwise on the retaining structure **121**. A bracket or an angled section **132** may be coupled to the side walls **127** to support the one or more battery modules **110** within the retaining structure **121**. The upper wall **126** extending therebetween may have a similar width as upper ends of the side walls **127** to which it is coupled.

[0045] The retaining structure **121** removably retains or secures the one or more battery modules **110** therein and may provide a support to secure the battery unit **100** to the aircraft. In an embodiment, attachment points and/or lugs (not shown) may be coupled to the retaining structure **121** for securing the battery unit **100** to the aircraft via tie down rods (not shown) attached to the aircraft. The tie down rods secure the battery unit **100** by applying a restraining force at the attachment points and/or lugs on the retaining structure **121**.

[0046] In the embodiment shown in FIG. 3, strap connectors **133** are connectable to straps and/or handles **136** coupled to the retaining structure **121** for carrying or moving the battery unit **100**.

[0047] Thermal runaway is a risk related to use of battery cells **120**, including lithium-ion battery cells, and a battery module **110** having a monoblock **115** that is damaged due to thermal runaway of the battery cells **120** may be required to be replaced. A thermal event may cause an increase in temperature and/or pressure in the battery monoblock **115** that may result in thermal runaway. For example, a short circuit in a battery cell **120** may generate excess heat within a battery monoblock **115** that may begin a chain of exothermic reactions that results in a rise in a temperature and/or pressure of the battery cells **120** of the battery monoblock **115**.

[0048] Various methods of venting and thermally isolating each battery monoblock **115** to prevent a thermal event in one monoblock **115**, or to prevent a thermal event in one monoblock **115** from effecting a thermal event in another monoblock **115** of the battery unit **100**, are available. In an

embodiment, a vent manifold system **134** coupled to each of the battery monoblocks **115** of the battery modules **110** is configured to allow venting of effluent gasses released from each battery monoblock **115** to the environment such that the other battery modules **110** in the battery unit **100** remain unaffected. U.S. Provisional Patent Application No. 63/496,789 to Henderson et al. (“Henderson et al.”), incorporated herein in its entirety by reference, discloses a vent manifold configured such that effluent gases produced in one or more battery monoblocks of a battery unit are discharged to the vent manifold and then from the vent manifold to the atmosphere to prevent development of heat and/or pressure that might result in propagation of thermal runaway to adjacent monoblocks and degradation or combustion of battery cells in the adjacent monoblocks. In Henderson et al., a plurality of battery cells each having a vent or pressure relief rupture disc are contained in a battery monoblock. Each monoblock includes a ventilation chamber or monoblock enclosure comprising a cell retaining structure or cell holder or a plurality of receptacles configured to receive the cells such that the vents of the cells are positioned to vent the gases within the ventilation chamber. The ventilation chamber is configured to contain gases vented by the cells within each battery monoblock and direct or release the gases in each monoblock through an opening or aperture, which may be covered by a rupture disc or an adhesive film or a one-way actuatable valve, to the vent manifold when a pressure threshold and/or temperature threshold within the monoblock is exceeded.

[0049] In an embodiment, the vent manifold system **134** is shared between the battery monoblocks **115** of the battery unit **100** and a valve or membrane (not shown), such as an independently actuatable one-way valve, extends between a ventilation chamber for each of the monoblocks **115** and the vent manifold system **134**. In an embodiment, each battery monoblock **115** includes an independently actuatable valve or membrane extending into the vent manifold system **134**. In an embodiment, the vent manifold system **134** includes one or more independently actuatable valves or membranes that extend into a corresponding battery monoblock **115** of the battery unit **100**. The vent manifold system **134** is configured to exhaust the gases from the one or more of the battery monoblocks **115** that have undergone or are about to undergo thermal runaway through an exhaust vent to the environment or an exterior of the battery unit **100** to prevent thermal runaway from occurring in other monoblocks **115** of the battery unit **100** or contaminating the other monoblocks **115** within the battery unit **100**. Each battery monoblock **115** and/or the vent manifold system **134** may also include a collection container (not shown) to prevent condensate that may collect due to a thermal runaway event from contaminating or causing damage, such as corrosion of components, to the adjacent monoblocks **115**.

[0050] The vent manifold system **134** may also be a structural member configured to provide structural support, as shown in FIG. 7, for removably securing the one or more battery modules **110** within the chassis or retaining structure **121** and/or for securing the battery unit **100** to the aircraft. In an embodiment, the vent manifold system **134** includes lugs **135** extending therefrom that engage tie down rods **137** attached to the aircraft to restrain and prevent movement of the battery unit **100**. As described herein, it is contemplated that the tie down rods **137** may be also secured to the retaining structure **121**.

[0051] It is foreseeable that the battery unit **100** may not include the vent manifold system **134** or may include a manifold that is not vented. An alternate means may be used to prevent a thermal runaway event from occurring in a monoblock **115** or from advancing to an adjacent monoblock **115**.

[0052] In an embodiment described herein, the battery unit **100** includes one or more battery modules **110** that may be efficiently and safely replaced, removed, or inserted, such as, for example, if a battery module **110** is faulty or inefficient and needs replacement or a thermal event has damaged the monoblock **115**, or if the number of battery modules **110** in the battery unit **100** needs to be selectively reduced or increased, such as, for example, to decrease a weight of the battery unit **100** or to increase an electrical capacity provided to the aircraft systems. In an

embodiment, each battery module **110** is configured to be removed and/or replaced, or is hot-swappable such that the battery module **110** may be removed and/or inserted in a process that does not include powering off the battery module **110** or the aircraft systems prior to mounting or dismounting the battery module **110** in the battery unit **100**. U.S. patent application Ser. No. 18/745,643 to Tidball et al., incorporated herein in its entirety by reference, includes a battery unit having one or more battery modules that each include a battery monoblock electrically connected to a high-power path that extends through or is coupled to a solid-state relay (SSR). The high-power path provides electrical power to a powered circuitry system of the battery module and to the aircraft systems. The SSR is triggered to disengage the high-power path, disengaging the terminals of the battery module, when the battery module is dismounted from the battery unit. In addition, a low-power path provides electrical power to the powered circuitry system when the high-power path is disconnected from the powered circuitry system. The high- and low-power paths and the SSR enable at least the following: hot swappability of the battery module when dismounting from or mounting to the battery unit, activation of the low-power path to enable functionality of electronics in the battery module when the high-power path is disengaged, and prevention of potential risks associated with the high-power path terminals remaining live when the battery module is removed from the retaining structure.

[0053] In an embodiment shown in FIG. **8**, the battery module **110**, which may be a hot-swap battery or not, includes a monoblock **115** containing one or more of the battery cells **120** and a control system or control electronics **310** removably coupled to the monoblock **115**. The control system **310** includes a solid-state relay printed circuit board (“SSR PCB”) **138** and a control system printed circuit board (“control PCB”) **140**. The SSR PCB **138** includes an electronic switch **163** configured to turn on and turn off the power being supplied to the aircraft from the battery module **110**. The electronic switch **163** is actuated by the SSR PCB **138** that is controlled by the control PCB **140**.

[0054] The control PCB **140** is electrically and communicatively connected between the SSR PCB **138** and the monitoring system PCB **122**. As described herein, the control PCB **140** receives signals from the monitoring system PCB **122** and the monoblock **115** and transmits a control signal to control the electronic switch **163** of the SSR PCB **138** via a control circuit **189**. In the embodiment shown, an electrical connection between the monoblock **115** and the electrical power distribution system of the aircraft is controlled by a monoblock powered circuitry system **165** that includes the control system **310** and the monitoring system PCB **122**.

[0055] The control system **310** includes functions for monitoring, controlling, and protecting various aspects of the battery cells **120** in the monoblock **115**. The control system **310** of the battery unit **100** is communicatively coupled to the monitoring system PCB **122** of the retaining structure **121** and transmits data to and from the monitoring system PCB **122** through a communication path **350**. Such data may include information regarding temperature, voltage, and/or current drawn from the monoblock **115**. The control system **310** is also communicably coupled to the aircraft and transmits data to and from the aircraft, or another user, through the monitoring system PCB **122** and via an aircraft signal connector **157**. Such transmittals may be for diagnostics and include data or information regarding a status of each battery module **110** and/or the battery unit **100**, including information regarding dismounting or disconnection of a battery module **110**, a capacity of each battery module **110** and/or the battery unit **100**, which battery modules **110** are available for energy transference, and the health and status of the battery modules **110** and the battery unit **100**. Such communication provides information that may be used to coordinate system energy usage and facilitate power usage prioritization. Further, data from each battery module **110** may be processed by the monitoring system PCB **122**, and the processed data, such as, for example, an average or representative battery temperature, number and identification of faulted modules or monoblocks, and overall battery health status, is transmitted from the monitoring system PCB **122** of the battery unit **100** to the aircraft avionics system via the aircraft signal connector **157**. In one embodiment,

data regarding the capacity, temperature, and/or pressure of each battery module **110** is transmitted to the aircraft avionics system via the aircraft signal connector **157**.

[0056] The control system **310** comprising the SSR PCB **138** and the control PCB **140** includes a control system enclosure **320**, shown in FIG. **4**, which protectively and/or thermally insulates and encloses the control system **310**. The control system enclosure **320** comprises a control system enclosure lid **147**, a control system enclosure base **325**, and various gaskets and fasteners that are configured to seal the control system **310** therein. In an embodiment, the control system **310** is removably coupled to the monoblock **115** to facilitate removal and replacement of the monoblock **115** from the battery module **110** such that the control system **310** can be coupled to a second monoblock **115** for use. In an embodiment, a snubber printed circuit board (“snubber PCB”) **330** may also be included in the control system **310** for battery module **110** protection and performance enhancement and may be coupled electrically in parallel with the SSR PCB **138**. In an embodiment, the SSR PCB **138**, the control PCB **140**, and/or the snubber PCB **330** may be combined into a single PCB.

[0057] The control system enclosure **320** containing the control system **310** is secured to a forward end **340** of the battery module **110**, such that the associated SSR PCB **138**, control PCB **140**, and/or snubber PCB **330** contained therein are thermally isolated and any electrolyte gases or other gases and/or liquids exiting or being vented from the associated monoblock **115** are prevented from entering the control system enclosure **320** to contaminate and/or damage the control system **310**. The control system enclosure **320** comprises one or more of a combination of materials having high electrical conductivity, high environmental resistance, and resistance to corrosion from vented electrolyte gases and/or other gases that may be produced and released from the monoblock **115**. As described herein, the electrolyte gases and/or other gases produced within the monoblock **115** during a thermal runaway event are released through the opening in the monoblock enclosure of the battery module **110** to the vent manifold system **134** coupled thereto, which vents the gases to the environment.

[0058] The control PCB **140** of the control system **310** is electrically coupled to a control circuit or control power path **150** and communicatively coupled to the communication path **350**, both of which engage the monitoring system PCB **122** that is housed in the retaining structure **121**. As shown in FIGS. **3** and **8**, the control power path **150** and the communication path **350** are coupled to and extend from the control PCB **140** of the control system **310** via a connector **185** that is engageable with the monitoring system PCB **122**. Each battery module **110** is removably coupled to the monitoring system PCB **122** of the retaining structure **121** via the connector **185**.

[0059] Within the battery module **110**, the control PCB **140** may be communicatively coupled to the battery cells **120** via a cell balancing and voltage sense path or communication path **360**, which may comprise one or more of charge balancing systems for measuring and balancing or equalizing the voltages and states of charge of the battery cells **120** in the battery monoblock **115**. The communication path **360** may also be used as a means of communicating a control signal between the monoblock **115** and the control PCB **140** to control the SSR PCB **138**. Further, the control PCB **140** may comprise a control circuit having a current sensor **196** that transmits a voltage signal to the control PCB **140** proportional to a current through the SSR PCB **138** from a shunt (not shown) embedded within the SSR PCB **138**. This information is used to monitor abnormal current conditions, such as, for example, a short circuit in either direction (into or out of the monoblock **115**), and for calculating, for example, a state of charge of the monoblock **115**, etc. that is communicable with the monitoring system PCB **122**.

[0060] The control system **310** may comprise a temperature sensing device or a means for measuring a temperature of the associated battery module **110**, such as, for example, a resistance temperature detector (RTD), a thermocouple, a negative temperature coefficient thermistor, a semiconductor-based temperature sensor, an analog device that reads an analog voltage, or another temperature sensing device known in the art. Such information may be communicated to at least

the control PCB **140**, and the control PCB **140** may engage one or more embedded heaters coupled to tabs or current collectors incorporated within the battery module **110**. Further, such information may be communicated to the monitoring system PCB **122** for processing and/or transmittal to aircraft avionics. In an embodiment, each battery module **110** includes at least one RTD (not shown). The battery unit **100** may also include at least one RTD positioned outside of the battery modules **110** for measuring a temperature of the battery unit **100**. The temperature measuring device may be directly coupled to the embedded heaters such that the embedded heaters are turned on or engaged when a first temperature is reached or turned off or disengaged when a second temperature is attained. The first temperature and the second temperature may be calculated or programmed temperatures. Data from the temperature sensing device may be used in assessing the health of each of the battery modules **110**.

[0061] The control system **310** may comprise a fault management system for monitoring defined fault criteria and identifying potential failure conditions and notifying aircraft avionics via the monitoring system PCB **122** and the aircraft signal connector **157** that fault criteria thresholds have been or are approaching being met. The control system **310** may include processes or programming known in the art for rectifying potential issues. In an embodiment, the fault management system of the control system **310** rectifies potential issues by electrically disconnecting the battery module **110** from the battery unit **100**. Further, the control system **310** may include passive components, such as fuses, to protect the battery module **110** against overvoltage and overload. It is foreseeable that the monitoring system PCB **122** may comprise a fault management system for monitoring fault criteria.

[0062] The battery unit **100** is arranged to enable protection of the monitoring system PCB **122** from potential thermal events. As best shown in FIG. 3, the monitoring system PCB **122** and an electrical bus system **153** are positioned between the terminal wall **123** and the partition wall or removable panel **151**. In an embodiment, the electrical bus system **153** is coupled to the removable panel **151**, and the removable panel **151** extends between the monitoring system PCB **122** and the battery modules **110**. The removable panel **151** may provide isolation of the monitoring system PCB **122** and/or other electronics that are adjacent the terminal wall **123** from the battery modules **110**. The battery modules **110** are positioned in the retaining structure **121** in engagement with the removable panel **151**, and each battery module **110** electrically and/or communicatively engages the electrical bus system **153** and the monitoring system PCB **122**. In an embodiment, the control system **310** of each battery module **110** is positioned to engage the monitoring system PCB **122** as described herein.

[0063] The battery modules **110** each include a high power path circuit that includes high-power paths **148** and **149** that are in engagement with the electrical bus system **153** via apertures or terminals **170** and **188**, respectively, in the removable panel **151**. The electric bus system **153** includes bus bars that provide an electrical junction or connection for conducting electricity from the one or more battery modules **110** to a battery power terminal connector **154** of the battery unit **100**. An aircraft power connector **370** is couplable to the battery power terminal connector **154** of the battery unit **100** to enable an electrical connection of the high-power paths **148** and **149** of the battery unit **100** to the electrical power distribution system of the aircraft.

[0064] The battery modules **110** each include the connector **185** that includes the control power path **150** and the communication path **350** from the control PCB **140** of the control system **310**. The connector **185** extends through an aperture **158** (see FIG. 2) in the removable panel **151** to engage the monitoring system PCB **122**. In an embodiment, the control power path **150** that extends between or is coupled between the control PCB **140** and the monitoring system PCB **122** comprises the low-power paths **400** and **410**, as shown in FIG. 8.

[0065] The high-power paths **148** and **149** and the low-power paths **400** and **410** of each of the monoblocks **115** of the battery unit **100** are electrically coupled to the monitoring system PCB **122** such that the monitoring system PCB **122** may be powered by one or more of the monoblocks **115**.

In the embodiment shown, the positive high-power path **148** is connected to a positive terminal of the monoblock **115** and is connected to the electric bus system **153** of the removable panel **151** via the terminal or connector **170**. The high-power return path **149** is connected to a negative terminal of the monoblock **115** and is connected to the electrical bus system **153** via the terminal or connector **188**. The electrical bus system **153** engages or is electrically connected to the monitoring system PCB **122**. The positive low-power path **400** is connected to a positive terminal of the monoblock **115**, and the low-power return path **410** is connected to a negative terminal of the monoblock **115**. The low-power paths **400** and **410** may form the control power path **150** connected to the monitoring system PCB **122** via the connector **185**.

[0066] A switch **215** (e.g., push button switch) shown in FIGS. **1** and **8** provides a means to selectively engage the low-power paths **400** and **410** for various functions, such as when the high-power paths **148** and **149** are disconnected from the powered circuitry system. The functions of the switch **215** may include, for example, engaging the control system **310** to provide a means to measure a voltage of each battery module **110**; providing electrical power to the monitoring system PCB **122** and the control system **310** of a battery module **110** to enable activation of one or more embedded heaters for preheating the battery cells **120** in the monoblock **115**; and providing power such that the monitoring system PCB **122** and the control system **310** are able to initiate, boot, and/or control engagement of the high-power path **148**.

[0067] In an embodiment, the removable panel **151** is not required to withstand high temperatures and/or high pressures that may exist during a thermal runaway event because each monoblock **115** and the vent manifold system **134** cooperate to protect the rest of the battery unit **100** from the effects of thermal events and/or thermal runaway. In an embodiment, the removable panel **151** is a mechanical cover that functions to isolate and protect the monitoring system PCB **122** and may be formed from one or a combination of materials having a high temperature resistance to thermally insulate the monitoring system PCB **122** from high temperatures due to activity in the battery modules **110** that release heat.

[0068] The monitoring system PCB **122** is installed on or adjacent to the terminal wall **123** of retaining structure **121**. As described herein with respect to the control PCB **140**, the monitoring system PCB **122** is electrically coupled to the control system **310** of each battery module **110** via a control power path **150** such that the control system **310** of each battery module **110** is powered through the monitoring system PCB **122**, and the monitoring system PCB **122** is communicatively coupled to the control system **310** via the communication path **350**. The control power path **150** and the communication path **350** couple the control PCB **140** to the monitoring system PCB **122** via the connector **185**. The monitoring system PCB **122** is electrically connected to the monoblock **115** via the high-power paths **148** and **149** and, additionally or alternatively through the low-power paths **400** and **410**. In an embodiment, the monitoring system PCB **122** detects the presence or absence of a voltage from the monoblock **115** and communicates such to the control system **310**. In an embodiment, communications or commands to and from the control system **310** is the presence or absence of a voltage.

[0069] The monitoring system PCB **122** may record, via the control system **310**, an identification or serial number associated with each battery module **110** and/or battery monoblock **115** connected to the monitoring system PCB **122** and fault events, such as, for example, short circuit faults, associated with each serial number. Such recording is to prevent a monoblock **115** from being reused if a non-resettable fault occurs. The monitoring system PCB **122** is programmable to reject a monoblock **115** having a non-resettable fault event previously recorded by the monitoring system PCB **122**. In an embodiment, the monoblock **115** is also able to store data related to fault events.

[0070] In an embodiment, the monitoring system PCB **122** includes the aircraft signal connector **157** that is connectable to the aircraft to communicate information or data or a status of the battery unit **100**, including when one or more of the battery modules **110** are dismounted. One or more of the battery modules **110** may provide electrical power to the monitoring system PCB **122** via the

aircraft signal connector **157** through at least the low-power paths **400** and **410** to enable the monitoring system PCB **122**. As shown in FIG. **8**, a connector jumper **194** of the aircraft signal connector **157** electrically connects the low-power paths **400** and **410** of a battery module **110** to the monitoring system PCB **122**. The connector jumper **194** may be implemented elsewhere in the aircraft, so long as the signal is fed back to the aircraft signal connector **157**.

[0071] The monitoring system PCB **122** communicates with each battery module **110** and with aircraft avionics. The monitoring system PCB **122** may include its own fault management system that is able to send commands to the control system **310** of one or more of the battery modules **110** to disconnect at least one of the monoblocks **115** from the high-power path **148**. When the monitoring system PCB **122** sends disconnection commands to all battery modules, the battery unit **100** is electrically isolated from the electrical power distribution system of the aircraft, but the aircraft signal connector **157** may still be engaged to receive information from the battery unit **100**. [0072] In an embodiment of the battery unit **100**, the one or more battery modules **110** comprise the high-power path **148** extending from or coupled to the positive terminal of the battery cells **120**, and the high-power path **148** is coupled to the SSR PCB **138** which has an electronic switch **163**. The high-power path **148** provides electrical power from the monoblock **115** via the high-power positive terminal **170** and an electrical bus system high-power path **168** to the battery power terminal connector **154** which is coupled to the aircraft power connector **370** to provide power to the electrical power distribution system of the aircraft, and to the monoblock powered circuitry system **165** when the electronic switch **163** of the SSR PCB **138** is engaged or switched “on” in response to an input voltage detected by the control system **310**.

[0073] The low-power paths **400** and **410** contained within the connector **185** are part of an alternate, current-limited circuit that, in an embodiment, may extend from or be coupled to the positive and negative terminals, respectively, of the battery cells **120**. The low-power paths **400** and **410** provide power to the monitoring system PCB **122** and the control system **310** when the electronic switch **163** of the SSR PCB **138** is disengaged or switched “off.” In an embodiment, the low-power paths **400** and **410** provide a limited electrical current of less than five milliamperes. In another embodiment the low-power paths **400** and **410** provide less than three milliamperes. In yet another embodiment the low-power paths **400** and **410** provide less than two milliamperes from the monoblock **115**. The low-power paths **400** and **410** provide power to at least the monitoring system PCB **122** and the control system **310**, and may have one or more functions, including, for example: providing a means to measure a voltage of the monoblock **115**, such as, for example, by terminals or test pins **167** extending within the aircraft signal connector **157** and denoted in FIG. **8** by VTest (+) and VTest (−); to enable the one or more embedded heaters for heating the monoblock **115**; and to provide power such that the monoblock powered circuitry system **165** is able to initiate, boot, and control engagement of the high-power paths **148** and **149**.

[0074] When the battery module **110** is mounted on the retaining structure **121**, but the monoblock **115** is electrically disengaged from the monitoring system PCB **122**, the switch **215** may be selectively actuated, providing initialization power to the monitoring system PCB **122** via the low power paths **400** and **410**. The monitoring system PCB **122** may, after a selected length of time, provide power to the control system **310**, which in turn may actuate the electronic switch **163** of the SSR PCB **138** coupled to the high-power path **148**, engaging the monoblock **115** to provide power to the aircraft and provide a primary power path to the monitoring system PCB **122** that, subsequently, powers the control system **310**.

[0075] When the battery module **110** is dismounted, the connector **185**, which includes the control power path **150** (i.e. the low power paths **400** and **410**) and the communication path **350**, extending through the aperture **158** in the removable panel **151** to engage the monitoring system PCB **122** in the retaining structure **121** is disconnected prior to the connectors **170** and **188** being disconnected, which include the high-power paths **148** and **149**, respectively. Similarly, upon mounting, the connectors **170** and **188** are connected first, and the connector **185** is subsequently connected. For

connector **185**, this can be referred to as a last make/first break (“LMFB”) connection. The LMFB connection of connector **185** may enable the control system **310** to establish that the battery module **110** is being dismounted or mounted. In an embodiment, when the battery module **110** is dismounted, the low-power paths **400** and **410** do not provide the alternate source of electrical power through the aircraft signal connector **157** to the monoblock powered circuitry system **165**, confirming that the battery module **110** has been dismounted from the retaining structure **121**. In an embodiment, the control system **310** senses and determines, from the sequence of electrical disconnects of the power paths and/or whether the low-power paths **400** and **410** transmit power through the connector jumper **194** to monitoring system PCB **122**, whether the battery module **110** is being dismounted or whether the battery module **110** is electrically disconnected (the electronic switch **163** coupled to the high-power path **148** is opened or disengaged) but the battery module **110** remains mounted.

[0076] Many different arrangements of the various components depicted, as well as components not shown, are possible without departing from the spirit and scope of the present disclosure. Embodiments of the present disclosure have been described with the intent to be illustrative rather than restrictive. Alternative embodiments will become apparent to those skilled in the art that do not depart from its scope. A skilled artisan may develop alternative means of implementing the aforementioned improvements without departing from the scope of the present disclosure. It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations and are contemplated within the scope of the claims.

Claims

1. A battery unit comprising: one or more battery modules, each battery module comprising a monoblock and a control system coupled to the monoblock, the control system actuatable to selectively engage or disengage at least one first circuit coupled to the monoblock of each of the one or more battery modules; a retaining structure configured to contain the one or more battery modules and including a power connector that provides power from the one or more battery modules; a monitoring system that is in communication with the control system of each of the one or more battery modules; wherein: the at least one first circuit couples each monoblock of the one or more battery modules to the power connector and the monitoring system, each of the one or more battery modules is removable from the retaining structure by disconnecting at least the at least one first circuit and a communication path that couple the control system and the monitoring system, and the power connector provides power when the at least one first circuit of at least one of the one or more battery modules is engaged.
2. The battery unit of claim 1, further comprising a second circuit that couples the monoblock of each of the one or more battery modules to at least the monitoring system, and each of the one or more battery modules is removable from the retaining structure by further disconnecting a control power circuit that comprises the second circuit and that couples the control system of each of the one or more battery modules to the monitoring system.
3. The battery unit of claim 1, further comprising a second circuit that couples the monoblock of each of the one or more battery modules to at least the monitoring system, the second circuit comprising an aircraft signal connector that is configured to communicatively connect the monitoring system to an aircraft.
4. The battery unit of claim 1, wherein the control system is housed within an enclosure and the control system is thermally isolated from the monoblock.
5. The battery unit of claim 1, wherein the monoblock is removable from the control system.
6. The battery unit of claim 1, further comprising a manifold coupled to the one or more battery modules, the manifold configured to receive effluent gases produced within the monoblock of each

of the one or more battery modules.

7. The battery unit of claim 1, wherein the control system of each of the one or more battery modules includes a temperature sensing device for sensing a temperature of the monoblock thereof, and the control system transmits data from the temperature sensing device to the monitoring system.

8. The battery unit of claim 1, wherein the monitoring system includes a fault management system that is in communication with the control system of each of the one or more battery modules to actuate the control system to selectively engage or disengage the at least one first circuit.

9. The battery unit of claim 8, wherein the fault management system stores data related to at least non-resettable fault events of each monoblock of the one or more battery modules retained in the retaining structure, and the monitoring system is in communication with the control system of each of the one or more battery modules to disengage the monoblock thereof having data stored in the fault management system related to a non-resettable fault event.

10. The battery unit of claim 1, further comprising a second circuit coupled to the monoblock of each of the one or more battery modules that includes means to test a voltage of the monoblock connected thereto.

11. A battery unit configured to provide electrical power to an aircraft, the battery unit comprising: a retaining structure; a monitoring system; one or more battery modules retainable in the retaining structure, wherein each battery module comprises: a monoblock comprising one or more battery cells, a first circuit that is coupled to the monoblock and engages at least the retaining structure and monitoring system, a second circuit that is coupled to the monoblock and engages at least the monitoring system, the second circuit comprising an aircraft signal connector that is configured to communicatively connect the monitoring system to the aircraft, and a control system that is actuatable to selectively engage or disengage the first circuit, wherein the control system is housed within an enclosure and is insulated from at least the monoblock of the battery module; wherein the control system of each of the one or more battery modules is in communication with the monitoring system via at least a communication path, and the one or more battery modules are each removable from the retaining structure by disconnecting at least the first circuit, the second circuit, and the communication path.

12. The battery unit of claim 11, wherein the monoblock is removable from the control system.

13. The battery unit of claim 11, wherein the monoblock of each of the one or more battery modules includes a chamber configured to contain effluent gases produced by the one or more battery cells, and the battery unit further comprises a manifold coupled to the chamber of each of the one or more battery modules configured to receive effluent gases contained in the chamber of each monoblock.

14. The battery unit of claim 11, wherein the control system of each of the one or more battery modules includes a temperature sensing device for sensing a temperature of the monoblock thereof, and the control system transmits data from the temperature sensing device to the monitoring system.

15. The battery unit of claim 11, wherein the monitoring system includes a fault management system that is in communication with the control system of each of the one or more battery modules to actuate the control system to selectively engage or disengage the first circuit.

16. A battery unit configured to provide electrical power to an aircraft, the battery unit comprising: a retaining structure having a power connector that is configured to be electrically connected to the aircraft; a monitoring system positioned within the retaining structure; one or more battery modules retained in the retaining structure and coupled to at least the monitoring system, wherein each battery module comprises: a monoblock comprising one or more battery cells, a high power path circuit that is coupled to the monoblock and engages the power connector and the monitoring system, a low power path circuit that is coupled to the monoblock and engages at least the monitoring system at least when the high power path circuit of the battery module is disengaged,

wherein the low power path circuit is a current-limited circuit, and a control system coupled to the monoblock and communicatively connected to the monitoring system, wherein the control system controls an electronic switch that is actuatable to selectively engage or disengage the high power path circuit, and the control system is housed within an enclosure and is insulated from at least the monoblock of the battery module; wherein the power connector of the retaining structure is configured to be electrically connected to the aircraft when the high power path circuit of at least one of the one or more battery modules is engaged, and the one or more battery modules are each removable from the retaining structure by disconnecting at least the high power path circuit, the low power path circuit, and a communication path between the control system of the one or more battery modules and the monitoring system.

17. The battery unit of claim 16, wherein the monoblock is removable from the control system.

18. The battery unit of claim 16, wherein the monoblock of each of the one or more battery modules includes a chamber configured to contain effluent gases produced by the one or more battery cells, and the battery unit further comprises a manifold coupled to the chamber of each of the one or more battery modules configured to receive effluent gases contained in the chamber of each monoblock.

19. The battery unit of claim 16, wherein the control system of each of the one or more battery modules includes a temperature sensing device for sensing a temperature of the monoblock thereof, and the control system transmits data from the temperature sensing device to the monitoring system.

20. The battery unit of claim 16, wherein the monitoring system includes a fault management system that is in communication with the control system of each of the one or more battery modules to actuate the control system to selectively engage or disengage the high power path circuit that electrically connects the monoblock to at least the power connector and the monitoring system.

21. The battery unit of claim 16, wherein the low power path circuit comprises an aircraft signal connector that is configured to communicatively connect the monitoring system to the aircraft.
